

Yingrui Hou

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10 September 2026

Mr Marc Sorel
Informatis-TS
26 rue Daubenton
75005 Paris, France

Application - Data Scientist / AI Engineer - Reference 3214

Dear Mr Sorel,

I am applying for the Data Scientist / AI Engineer position in Paris or Clermont-Ferrand. The opportunity to explore new use cases, develop proofs of concept and help teams assess technical feasibility is particularly appealing to me. Based in Clermont-Ferrand, I would welcome the opportunity to bring my experience in scientific data analysis and model validation to your client's innovation team.

I hold a PhD in particle physics and have four years of postdoctoral research experience at CNRS. At LAPP, I developed Python and C++ workflows for machine learning, statistical inference and validation on complex measurement data. This included feature studies, train/test diagnostics and sample reweighting to address differences between datasets. I also developed monitoring methods that traced discrepancies to calibration and software causes, including a reconstruction issue that was subsequently corrected.

At LPCA, my work connected experimental measurements with simulation to evaluate the behaviour of an instrumented prototype. I developed calibration, response-mapping and signal-analysis workflows, then used their outputs to assess model assumptions and performance limits. In a separate collaborative study, I contributed to proof-of-principle validation of a new inference method under realistic measurement effects. These projects required me to turn open questions into testable approaches and communicate what the evidence did and did not support.

I would bring scientific Python skills, rigorous machine learning validation and experience developing reusable analytical components in shared Linux and Git environments. My professional platform experience has mainly been in scientific computing rather than Azure or Databricks, so I would need to build practical familiarity with your client's environment. My immediate contribution would be in modelling, experimental design, validation and clear technical recommendations.

I work comfortably in English within international collaborations and value close discussion with domain experts when defining a useful analysis. I would be pleased to discuss how my background could support your client's exploratory projects and the transition from a tested idea to a documented solution.

Kind regards,
Yingrui Hou